Task 1 Report Content

Title

RTL Design and Functional Simulation of 8-Bit ALU Using Verilog

Objective

To design and verify an 8-bit Arithmetic Logic Unit (ALU) capable of performing arithmetic and logical operations using Verilog HDL and Vivado Simulator.

Operations Implemented

* Addition
* Subtraction
* AND
* OR
* XOR
* Left Shift
* Right Shift
* NOT

Tools Used

* Vivado 2025.2
* Verilog HDL
* XSim Simulator

Results

The ALU was successfully simulated. All eight operations produced the expected outputs. Functional verification was performed using a Verilog testbench, and waveform analysis confirmed correct behavior.

Conclusion

The 8-bit ALU was successfully designed, simulated, and verified. The RTL implementation met all functional requirements and demonstrated correct operation for all supported arithmetic and logical functions.